Unbiased Estimators & Confidence Intervals

ANSWER KEY

Point estimates for the mean and variance

1 A sample of 8 measurements has values $12, 15, 11, 14, 13, 16, 12, 15$. Find the unbiased point estimate of the population mean.

$\bar{x} = \frac{12+15+11+14+13+16+12+15}{8} = \frac{108}{8} = 13.5$. This sample mean is itself the unbiased estimate of the population mean μ.

2 Explain why the unbiased estimator of population variance divides the sum of squared deviations by $n-1$ rather than n, using the formula $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$.

Dividing by n would systematically underestimate the true population variance on average, because $\bar{x}$ is calculated FROM the same sample, making the sample's squared deviations from $\bar{x}$ slightly smaller than deviations from the true (unknown) population mean μ would be. Dividing by $n-1$ (the degrees of freedom) corrects this bias, making s^2 an unbiased estimator of σ^2.

Computing the unbiased sample variance

3 For the sample $4, 7, 9, 6, 9$, find the unbiased estimate of the population variance.

Mean $= \frac{4+7+9+6+9}{5} = \frac{35}{5} = 7$. Squared deviations: $(4-7)^2 = 9$, $(7-7)^2 = 0$, $(9-7)^2 = 4$, $(6-7)^2 = 1$, $(9-7)^2 = 4$. Sum $= 18$. $s^2 = \frac{18}{5-1} = \frac{18}{4} = 4.5$.

Constructing a confidence interval for the mean

4 A sample of $n = 36$ has mean $\bar{x} = 52$ and population standard deviation $\sigma = 6$ (known). Construct a 95% confidence interval for the population mean, using $z^* = 1.96$.

Standard error: $\frac{\sigma}{\sqrt{n}} = \frac{6}{\sqrt{36}} = 1$. Margin of error: $1.96(1) = 1.96$. Confidence interval: 52 ± 1.96, or $(50.04, 53.96)$.

5 A sample of $n = 25$ has mean $\bar{x} = 80$ and unbiased sample standard deviation $s = 10$ (population standard deviation unknown, so a t-distribution is used). Construct a 90% confidence interval, using $t^* \approx 1.711$ (with $df = 24$).

Standard error: $\frac{s}{\sqrt{n}} = \frac{10}{\sqrt{25}} = 2$. Margin of error: $1.711(2) = 3.422$. Confidence interval: 80 ± 3.422, or approximately $(76.58, 83.42)$.

Interpreting a confidence interval

6 A 95% confidence interval for a population mean is calculated as $(48.2, 55.8)$. State the correct interpretation of this interval, and identify a common misinterpretation to avoid.

Correct interpretation: if this sampling and interval-construction process were repeated many times, approximately 95% of the resulting intervals would contain the true population mean. A common misinterpretation to avoid: it is NOT correct to say there is a 95% probability that the true mean lies within this specific interval — the true mean is a fixed (though unknown) value, not a random variable, so once the interval is calculated it either does or does not contain it.

Sample size and interval width

7 A researcher wants to halve the margin of error of a confidence interval for the mean, keeping the confidence level and σ fixed. Explain how the required sample size n must change, using the formula for margin of error $E = z^* \frac{\sigma}{\sqrt{n}}$.

Since E is inversely proportional to $\sqrt{n}$, halving E requires $\sqrt{n}$ to double, which means n itself must be multiplied by 4 (quadrupled) — a direct consequence of the square-root relationship between sample size and margin of error.

Determining the required sample size

8 A researcher wants a 95% confidence interval for a population mean with a margin of error no larger than 2, given a known population standard deviation $\sigma = 9$. Find the minimum required sample size, using $z^* = 1.96$ and rounding up to the nearest whole number.

$E = z^* \frac{\sigma}{\sqrt{n}}$, so $n = \left(\frac{z^* \sigma}{E}\right)^2 = \left(\frac{1.96(9)}{2}\right)^2 = (8.82)^2 \approx 77.79$. Rounding up (sample size must always round up to guarantee the target margin of error): $n = 78$.

Confidence intervals for a proportion

9 A survey of 200 randomly selected voters finds that 120 support a proposal. Construct a 95% confidence interval for the true population proportion, using $\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1-\hat{p})}{n}}$ with $z^* = 1.96$.

$\hat{p} = \frac{120}{200} = 0.6$. Standard error: $\sqrt{\frac{0.6(0.4)}{200}} = \sqrt{0.0012} \approx 0.0346$. Margin of error: $1.96(0.0346) \approx 0.0679$. Confidence interval: 0.6 ± 0.0679, or approximately $(0.532, 0.668)$.

Comparing pooled vs. unbiased variance estimates

10 Two independent samples give unbiased variance estimates $s_1^2 = 16$ (from $n_1 = 10$) and $s_2^2 = 25$ (from $n_2 = 15$). Find the pooled variance estimate, using $s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$.

$s_p^2 = \frac{(9)(16) + (14)(25)}{10 + 15 - 2} = \frac{144 + 350}{23} = \frac{494}{23} \approx 21.48$.

Effect of confidence level on interval width

11 Explain why a 99% confidence interval is always wider than a 95% confidence interval for the same sample data, referencing the critical value z^* used in each.

A higher confidence level requires a larger critical value z^* (approximately 2.576 for 99% versus 1.96 for 95%), since capturing the true parameter with greater certainty requires a wider net. Since the margin of error $E = z^* \frac{\sigma}{\sqrt{n}}$ is directly proportional to z^*, a larger z^* produces a wider confidence interval for the same sample data — there is an inherent tradeoff between confidence level and precision.

Constructing a confidence interval for a difference of means

12 Two independent samples give $\bar{x}_1 = 45$ (with $n_1 = 40$, $\sigma_1 = 8$) and $\bar{x}_2 = 40$ (with $n_2 = 50$, $\sigma_2 = 6$). Construct a 95% confidence interval for $\mu_1 - \mu_2$, using $z^* = 1.96$ and the standard error formula $\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$.

Point estimate: $\bar{x}_1 - \bar{x}_2 = 45 - 40 = 5$. Standard error: $\sqrt{\frac{64}{40} + \frac{36}{50}} = \sqrt{1.6 + 0.72} = \sqrt{2.32} \approx 1.523$. Margin of error: $1.96(1.523) \approx 2.99$. Confidence interval: 5 ± 2.99, or approximately $(2.01, 7.99)$.

Assessing whether a claimed population mean is plausible

13 A machine is supposed to fill bottles with 500ml of liquid. A sample of $n = 30$ bottles has mean $\bar{x} = 497$ ml and sample standard deviation $s = 8$ml. Construct a 95% confidence interval for the true mean fill volume (using $z^* \approx 1.96$ as an approximation for the t-distribution with $df = 29$), and use it to comment on whether the machine's target of 500ml is plausible.

Standard error: $\frac{s}{\sqrt{n}} = \frac{8}{\sqrt{30}} \approx 1.461$. Margin of error: $1.96(1.461) \approx 2.86$. Confidence interval: 497 ± 2.86, or approximately $(494.14, 499.86)$. Since the target value of 500ml lies outside this confidence interval, there is evidence at the 95% confidence level that the machine's true mean fill volume is below the target of 500ml.

14 Explain why increasing the sample size n, while keeping everything else fixed, makes a confidence interval narrower, referencing the standard error formula $\frac{\sigma}{\sqrt{n}}$.

The standard error $\frac{\sigma}{\sqrt{n}}$ decreases as n increases, since n appears in the denominator under a square root. A smaller standard error directly produces a smaller margin of error $E = z^* \frac{\sigma}{\sqrt{n}}$, and hence a narrower confidence interval — intuitively, a larger sample gives a more precise (less variable) estimate of the population mean.

Interpreting overlapping confidence intervals

15 Two independent 95% confidence intervals for population means are $(10, 14)$ and $(13, 17)$. Explain what can and cannot be concluded about whether the two population means are different, given that the intervals overlap.

Since the two confidence intervals overlap (both contain values between 13 and 14), it is plausible that the two population means could be equal, so we cannot conclude at the 95% confidence level that the means are different. However, overlapping confidence intervals do not definitively prove the means are equal either — a formal hypothesis test comparing the two means directly (e.g. a two-sample t-test) would be needed to draw a rigorous conclusion, since the overlap-based reasoning is only an informal, conservative check.